

Md. Mahmudul Hasan

Assistant Professor, Department of Electrical & Electronic Engineering (EEE)

Rajshahi University of Engineering & Technology (RUET), Rajshahi- 6204, Bangladesh

📍 Rajshahi- 6204, Bangladesh ✉ mahmud@eee.ruet.ac.bd ☎ +8801837540698 🌐 eee.ruet.ac.bd/mahmud

📖 Google Scholar Profile in mahmud-eee-ruet

Research Interests

- Multi-objective metaheuristic optimizations for controller tuning and energy management in renewable-integrated power systems.
- Disturbance observer-based control system design for renewable energy applications.
- Virtual inertia emulation and frequency stabilization techniques for converter-dominated microgrids.

Education

Rajshahi University of Engineering & Technology

April 2020 – December 2024

M.Sc. in Electrical & Electronic Engineering

- **CGPA:** 3.92/4.0
- **Thesis:** Development of a Hybrid Metaheuristic Optimization Based Control Platform for Three Phase Microgrid System
- **Coursework:** Engineering Analysis, Modern Control Theory, Multivariable Control Systems, Renewable Energy, Energy Conversion, Digital Circuit Design

Rajshahi University of Engineering & Technology

January 2015 – August 2019

B.Sc. in Electrical & Electronic Engineering

- **CGPA:** 3.89/4.0 (1st class 2nd among 120 students)
- **Thesis:** Artificial Neural Network Based Sensorless Field Oriented Control of Three phase Induction Motor
- **Major:** Power System

Academic Positions

Assistant Professor

Rajshahi- 6204, Bangladesh

Department of EEE, Rajshahi University of Engineering & Technology

January 2024 – Present

- **Taught Courses:** Measurement and Instrumentation, Computational Methods in Engineering, Renewable Energy, Power System Operation and Control
- **No. of Undergraduate Thesis Supervised:** 5
- **Additional Role:** Member Secretary, OBE Based Undergraduate Curriculum Preparation Committee

Lecturer

Rajshahi- 6204, Bangladesh

Department of EEE, Rajshahi University of Engineering & Technology

February 2021 – December 2023

- **Taught Courses:** Measurement and Instrumentation, Power Plant Engineering and Economy, Power System II, Power System Operation and Control
- **No. of Undergraduate Thesis Supervised:** 3
- **Additional Role:** Member, Undergraduate Curriculum Modification Committee

Lecturer

Dhaka- 1216, Bangladesh

Department of EEE, Prime University

October 2019 – January 2021

- **Taught Courses:** Electronics I, Electromagnetic Fields, Digital Electronics and Pulse Techniques, Electrical Machine, Renewable Energy, Power System Operation and Control

Publications

Journal Articles

- [1] **M. M. Hasan**, M. S. Rana, T. S. Joyeeta, and F. Tabassum, "Virtual inertia emulation in islanded microgrids using a time-series disturbance observer for frequency stabilization," *International Journal of Electrical Power Energy Systems*, vol. 176, p. 111 673, 2026. DOI: <https://doi.org/10.1016/j.ijepes.2026.111673> .
- [2] R. Hasan, **M. M. Hasan**, M. S. Rana, F. Tabassum, and M. N. B. Reza, "Disturbance observer-aided SSA-TID control for frequency stabilization in hybrid renewable power systems," *IET Renewable Power Generation*, vol. 19, no. 1, 2025. DOI: doi.org/10.1049/rpg2.70137 .
- [3] F. Tabassum, M. Rana, T. S. Joyeeta, and **M. M. Hasan**, "Optimal defense against security threat on stability of grid-integrated synchronous generator during fdi attack," *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 13, p. 101 097, 2025. DOI: [10.1016/j.prime.2025.101097](https://doi.org/10.1016/j.prime.2025.101097) .
- [4] M. N.-U.-Rahman, **M. M. Hasan**, A. M. Raka, *et al.*, "An asymmetric fuzzy-based self-tuned pso-optimized mppt controller for grid-connected solar photovoltaic system," *Energy Conversion and Management: X*, vol. 26, p. 100 902, 2025. DOI: [10.1016/j.ecmx.2025.100902](https://doi.org/10.1016/j.ecmx.2025.100902) .
- [5] **M. M. Hasan**, M. S. Rana, F. Tabassum, H. R. Pota, and M. H. K. Roni, "Optimizing the initial weights of a PID neural network controller for voltage stabilization of microgrids using a PEO-GA algorithm," *Applied Soft Computing*, vol. 147, p. 110 771, 2023. DOI: [10.1016/j.asoc.2023.110771](https://doi.org/10.1016/j.asoc.2023.110771) .
- [6] M. H. K. Roni, M. S. Rana, H. R. Pota, **M. M. Hasan**, and M. S. Hussain, "Recent trends in bio-inspired meta-heuristic optimization techniques in control applications for electrical systems: A review," *International Journal of Dynamics and Control*, vol. 10, no. 3, pp. 999–1011, Jan. 2022. DOI: [10.1007/s40435-021-00892-3](https://doi.org/10.1007/s40435-021-00892-3) .
- [7] M. R. Islam, J. Hasan, **M. M. Hasan**, M. N. Huda, M. A. H. Sadi, and A. AbuHussein, "Performance improvement of dfig-based wind farms using narma-l2 controlled bridge-type flux coupling non-superconducting fault current limiter," *IET Generation, Transmission and Distribution*, vol. 14, no. 26, pp. 6580–6593, 2021. DOI: [10.1049/iet-gtd.2019.1917](https://doi.org/10.1049/iet-gtd.2019.1917) .

Journal Articles Under Review

- [1] **M. M. Hasan**, M. M. Hasan, M. S. Rana, F. Tabassum, and T. S. Joyeeta, "Multivariable power control of grid-forming inverters using a PID neural network," Manuscript under review at *IET Renewable Power Generation*, 2025.

Selected Conference Papers

- [1] M. N. M. Rahi, **M. M. Hasan**, M. Y.-Y. U. Haque, A. I. Abeg, and T. S. Turna, "Voltage control of grid-forming inverter in islanded microgrids using PSO-optimized sliding mode controller," in *Proc. Int. Conf. Electr., Comput. Commun. Eng. (ECCE)*, 2025, pp. 1–6. DOI: [10.1109/ECCE64574.2025.11013018](https://doi.org/10.1109/ECCE64574.2025.11013018) .
- [2] M. A. Hossain, **M. M. Hasan**, F. Tabassum, and M. S. Rana, "Salp swarm algorithm based integral LQG control of two area nonlinear power system with renewable energy penetration," in *Proc. Int. Conf. Comput. Inf. Technol. (ICCIT)*, 2024, pp. 657–662. DOI: [10.1109/ICCIT64611.2024.11022531](https://doi.org/10.1109/ICCIT64611.2024.11022531) .
- [3] **M. M. Hasan**, M. S. Rahman, M. S. Rana, and F. Tabassum, "Optimal fractional order control of virtual inertia emulator for a microgrid with high renewable energy penetration," in *Proc. Int. Conf. Dev. Renew. Energy Technol. (ICDRET)*, 2024, pp. 1–6. DOI: [10.1109/ICDRET60388.2024.10504017](https://doi.org/10.1109/ICDRET60388.2024.10504017) .
- [4] M. A. Rafi, **M. M. Hasan**, M. Y.-Y. U. Haque, Z. Ferdous, M. S. Rana, and M. F. Hossain, "Performance augmentation of an arc suppression device using particle swarm optimization based proportional resonant controller," in *Proc. Int. Conf. Electr. Comput. Eng. (ICECE)*, 2022, pp. 445–448. DOI: [10.1109/ICECE57408.2022.10089057](https://doi.org/10.1109/ICECE57408.2022.10089057) .
- [5] **M. M. Hasan**, M. S. Hussain, M. S. Rana, and M. H. K. Roni, "Population extremal optimization based 2-DOF control strategy for field oriented control of induction motor," in *Proc. Int. Conf. Electr. Electron. Eng. (ICEEE)*, 2021, pp. 117–120. DOI: [10.1109/ICEEE54059.2021.9718799](https://doi.org/10.1109/ICEEE54059.2021.9718799) .
- [6] **M. M. Hasan** and M. S. Rana, "Modified population extremal optimization based PID neural network for islanded microgrid system," in *Proc. IEEE Int. Conf. Comput., Power Commun. Technol. (GUCON)*, 2021, pp. 1–6. DOI: [10.1109/GUCON50781.2021.9573922](https://doi.org/10.1109/GUCON50781.2021.9573922) .

- [7] **M. M. Hasan**, “Artificial neural network based speed estimator for sensorless field oriented control of three phase induction motor,” in *Proc. Int. Conf. Electr., Comput. Telecommun. Eng. (ICECTE)*, 2019, pp. 57–60. DOI: [10.1109/ICECTE48615.2019.9303559](https://doi.org/10.1109/ICECTE48615.2019.9303559) .

Supervised Undergraduate Thesis

1. *A 2-DOF Optimal Active Disturbance Rejection Framework for Load Frequency Control*, M H M Tawhidul Islam Fahim and Fayeze Naqib Efati, Department of EEE, RUET, 2025.
2. *Multi-Objective Optimization Based Energy Management System for Grid Connected Microgrid*, Susmita Paul, Department of EEE, RUET, 2025.
3. *Optimal Fractional Order Control of Virtual Inertia Emulator for a Microgrid with High Renewable Energy Penetration*, Md. Sohanur Rahman, Department of EEE, RUET, 2024.
4. *Salp Swarm Algorithm Tuned Fractional Order Controller for Load Frequency Control of Hybrid Renewable Power system*, Rakibul Hasan, Department of EEE, RUET, 2024.
5. *Optimal Active Disturbance Rejection Control of Power System Stabilizer*, Maliha Anam, Department of EEE, RUET, 2024.
6. *Robust Optimal Dynamic Voltage Restoration in a Compensated Resonant Grounded Distribution System*, Ramisa Fariha Mridha and Naimur Rahman, Department of EEE, RUET, 2023.
7. *Performance Augmentation of an Arc Suppression Device Using Active Disturbance Rejection Control*, Ragib Rowshon, Department of EEE, RUET, 2023.
8. *Voltage Control of PV-fed Grid Forming Inverter Based Three-Phase Microgrid Using Optimal Sliding Mode Controller*, Munsiri Nahid Md Rahi, Department of EEE, RUET, 2023.

Review Activities

- Reviewer, Electric Power Systems Research, Elsevier
- Reviewer, ISA Transactions, Elsevier
- Reviewer, International Transactions on Electrical Energy System, Hindawi
- Reviewer at different IEEE international conferences

Technical Skills

Optimization: Multi-objective metaheuristic optimizations, Convex optimization

Control System Design: State-feedback, observer based control, H_∞ control, Adaptive and neuro-adaptive control, disturbance observer design for linear and nonlinear systems, sliding mode control with observation

Deep Learning: Designing and training ANN, RNN, LSTM, and GRU

Programming: MATLAB, Python (PyTorch, Pandas, NumPy, Matplotlib), C

Simulation Tools: MATLAB/Simulink, Proteus, PSpice

Hardware Platforms: Arduino, Sensor integration, PCB design

Productivity Tools: LaTeX, Microsoft Office

UGC Funded Project

Project Title: Adaptive Control of Islanded Microgrid System using Population Extremal Optimization Based PID Neural Network

Role: Project Director

Grant No.: DRE/7/RUET/528(35)/PRO/2021-22/14

Duration: 2021-2022

Grant Amount: BDT 2,50,000

Industrial Training

Ghorashal Power Station

Bangladesh Power Development Board

Ghorashal, Bangladesh

08 May 2018 – 23 May 2018

- **Description:** The training was conducted on power generation (steam turbine, gas turbine, combined cycle, and gas engine power plants), substation, switchgear, and power plant simulator.

Standardized Test Score

IELTS Academic: Overall Band 8.0 (Listening: 9.0, Reading: 9.0, Writing: 6.5, Speaking: 7.0)

Professional Services

- Publication Secretary, IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), 12-13 September 2024. Venue: Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh.
- Publication Committee Member, IEEE 3rd International Conference on Electrical & Electronic Engineering (ICEEE), 22-24 December 2021. Venue: Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh.
- Promotional and Publication Sub-committee Member, 1st Reunion of EEE Alumni Association, RUET.
- Invited Talk: “Energy Management and Virtual Inertia Emulation in Microgrids”, IEEE PES Day, IEEE Student Branch, Bangladesh University of Business and Technology (BUBT), 2024.

Awards and Scholarships

- EEE Association Award for Academic Excellence, 2016
- Education Board Scholarship for Excellence in Higher Secondary School Certificate Examination, 2014
- Education Board Scholarship for Excellence in Secondary School Certificate Examination, 2012

References

Prof. Dr. S. M. Abdur Razzak
Professor, Department of EEE, RUET &
Vice-chancellor, RUET
Email: sma.razzak@eee.ruet.ac.bd
Relation: Undergraduate supervisor

Prof. Dr. Md Sohel Rana
Professor, Department of EEE, RUET
Email: sohel@eee.ruet.ac.bd
Relation: Postgraduate supervisor